

Mohamed Ben Mansour

AI Software Engineer | Multi-Agent Systems (MAS) | Backend Infrastructure

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Tunis, Tunisia

Professional Summary

AI Software Engineer specializing in autonomous Multi-Agent Systems (MAS) and production-grade AI infrastructure. Translating complex Agent-to-Agent (A2A) orchestration concepts into low-latency production pipelines using LangGraph and MCP. Gained expertise in integrating deep learning architectures into scalable backend ecosystems with strict CI/CD governance. Focused on building cost-optimized, highly observable enterprise AI automation.

Core Competencies

Agentic AI & Orchestration · Multi-Agent Systems (MAS) · LangGraph Workflows
Waterfall Data Enrichment · Asynchronous Backend APIs · Docker & CI/CD Governance
Enterprise AI Tools & MCP · Full-Stack Engineering · Human-in-the-Loop Systems

Professional Experience

AI Software Engineer Intern

Addvocate.ai

Feb 2026 – Present

Tunis, Tunisia

- **Architecture:** Architected an autonomous B2B outbound workflow via a Multi-Agent System (MAS), deploying specialized functional nodes (**Intelligence Sourcer, Strategic Qualifier, Personalization Engine**) to capture intent data, filter prospects, and generate context-driven engagement content.
- **Data Engineering:** Engineered local data-ingestion pipelines using **Neo4j** and a customized **waterfall enrichment algorithm**, optimizing operational infrastructure costs by ensuring local knowledge graph lookups and in-house scraping execute before cascading fallbacks to external data APIs.
- **Orchestration:** Designed stateful Agent-to-Agent (A2A) workflows via a centralized **Global Orchestrator** node using **LangGraph**, controlling cross-platform timing parameters while fully decoupling heavy enrichment cycles from live LLM reasoning windows.
- **Optimization:** Implemented a **Human-in-the-Loop (HITL)** state graph and automated feedback loops, dynamically updating personalization playbooks based on real-time corrections to minimize hallucinations, yielding a 3x increase in response rates and 40% higher lead conversion.

AI/DevOps Intern

Vermeg

June 2025 – July 2025

Tunis, Tunisia

- **Infrastructure:** Engineered containerized, Docker-based CI/CD deployment pipelines and structural SDLC governance scripts, minimizing manual configuration variations and decreasing deployment failures by **70%** based on production logs.
- **Observability:** Configured distributed infrastructure monitoring utilizing Prometheus and Grafana dashboards, integrating automated AI-driven log summarizations to execute root-cause analysis and reduce mean-time-to-debug (MTTD) by over **90%**.

Academic & Engineering Projects

Agentic MAS for Real-Time Fraud Detection

Independent AI Engineering Project

International Reply Challenge Hackathon

Online

- **Architecture:** Designed an autonomous Multi-Agent System (MAS) using **LangChain** to intercept financial fraud transactions, strictly separating deterministic Python-side evidence aggregation from LLM evaluations to scale token efficiency.
- **Algorithmic:** Engineered a structured ReAct execution model coordinating specific tool-subagents for geo-consistency tracking (Haversine formula), velocity evaluation, and metadata phishing analysis, limiting execution to a max of 2 LLM calls per transaction.
- **Telemetry:** Configured parallel execution workflows via a multi-threaded **ThreadPoolExecutor** structure for stateless concurrency, connecting **Langfuse** to capture live telemetry traces, execution costs, and internal token span counts.

EmbedAI – Intelligent Assistant for STM32 Development

AI Software Engineer (Academy-Industry Collaboration with ACTIA)

Feb 2025 – May 2025

Tunis, Tunisia

- **Core Dev:** Developed a context-aware embedded hardware copilot (Web application and VS Code extension) combining T5 architectures and custom Transformer models integrated with a FastAPI backend and a Clang deployment toolchain.
- **Automation:** Configured real-time programmatic compilation safety testing and automated feedback loops, cutting development debugging cycles and hardware verification times by **40%**.

Technical Skills

Languages & Core Infrastructure: Python, Java, C++, JavaScript, SQL, NoSQL (Neo4j), Git, Linux, Thread Pooling

AI Frameworks & Ecosystems: LangChain, LangGraph, MCP, A2A Orchestration, PyTorch, TensorFlow, OpenCV, Scikit-learn

Architecture & Deep Learning: Multi-Agent Systems, ReAct Framework, Transformers, Graph DBs, RAG, Model Optimization

Backend Engineering Systems: FastAPI, Django, Flask, Node.js, Spring Boot, RESTful APIs, Asynchronous Processing

DevOps & Observability Stack: Docker, CI/CD (Jenkins, GitHub Actions), Langfuse, Prometheus, Grafana

Human Languages: French (Proficient), English (Fluent)

Education

Esprit Engineering School – Engineering Degree in Artificial Intelligence Engineering

2024 – 2027

Faculty of Sciences of Tunis – Integrated Preparatory Cycle (Math, Physics, CS)

2022 – 2024

National Secondary Degree – Mathematics Baccalaureate, High Honors (Très Bien)

2022

Certifications

NVIDIA Deep Learning Institute:

- Building AI Agents with Multimodal Models (Nov 2025)
- Evaluation and Light Customization of Large Language Models (June 2025)
- Generative AI with Diffusion Models (Feb 2025)
- Fundamentals of Deep Learning (Dec 2024)